Unit 4: Further Mechanics, Fields and Particles

4.3 Further Mechanics

This topic covers impulse, conservation of momentum in two dimensions and circular motion.

It can be studied using applications that relate to, for example, a modern rail transportation system.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

Note that force is constant here

81	understand how to use the equation impulse $= F\Delta t = \Delta p$ (Newton's second law of motion)
	Impulse (Newton's second law of motion) Impulse (Ns) = Change in Momentum (kg m/s) = Force (N) x Time (s) $I = \Delta p = mv - mu = F\Delta t$
82	**CORE PRACTICAL 9: Investigate the relationship between the force exerted on an object and its change of momentum**
83	understand how to apply conservation of linear momentum to problems in two dimensions
	Conservation of Linear Momentum in 2-Dimensions The law of conservation of linear momentum states: ***"The total momentum before a collision = the total momentum after a collision provided no external force acts*** (i.e. in an isolated system)" - We can apply this law in two dimensions by treating the x and y components of motion separately - The **total momentum in the x-direction before an event must equal the total momentum in the x-direction after** (horizontal components must be equal before and after) - The **total momentum in the y-direction before an event must equal the total momentum in the y-direction after** (vertical components must be equal before and after) - This is because momentum is a vector and vectors can be broken down into perpendicular components (x and y) that are conserved independently
84	**CORE PRACTICAL 10: Use ICT to analyse collisions between small spheres, e.g. ball bearings on a table top**
85	understand how to determine whether a collision is elastic or inelastic
	Elastic & Inelastic Collisions - **Elastic collision** is a *collision in which both momentum and total kinetic energy is conserved* - Commonly seen in collisions where objects move away from each other after collision - **Inelastic collision** is a *collision in which momentum is conserved but the total kinetic energy is not conserved* - Commonly seen in collisiosn where objects stick together after collision !Note that only KE is taken into consideration in whether a collision is elastic or inelastic, *the total energy in all forms must always be conserved*

86 be able to derive and use the equation $E_k = \dfrac{p^2}{2m}$ for the kinetic energy of a non-relativistic particle

Derivation of KE = $p^2/2m$

$$E_k = \frac{1}{2}mv^2 \quad \text{and} \quad v = \frac{p}{m}$$

$$E_k = \frac{1}{2}mv^2$$

$$\therefore \quad E_k = \frac{1}{2}m\left(\frac{p}{m}\right)^2$$

$$\therefore \quad E_k = \frac{1}{2}\frac{p^2}{m}$$

$$\therefore \quad E_k = \frac{p^2}{2m}$$

This final formula is used in calculations involving the KE of subatomic particles **moving at *non-relativistic speeds*** (i.e. speed much slower than speed of light)

Extra knowledge (You do not need to know for your IAL studies)
As a relativistic particle's speed approaches the speed of light (c), its kinetic energy increases without bound, requiring an infinite amount of energy to reach c. The need for infinite energy means that an object with mass can never reach the speed of light and the equation for KE is no longer KE = ½ mv^2

87 be able to express angular displacement in radians and in degrees, and convert between these units

Radians and Degrees

- A **radian** is defined as:
  "The angle subtended at the centre of a circle by an arc equal in length to the radius of the circle"
- To convert from degrees to radians:

REARRANGE DEGREES TO RADIANS CONVERSION EQUATION

DEGREES ⟶ RADIANS $\quad \theta^\circ \times \dfrac{\pi}{180} = \theta$ RAD

RADIANS ⟶ DEGREES $\quad \theta$ RAD $\times \dfrac{180}{\pi} = \theta^\circ$

Angular displacement ($\Delta\theta$)

- **Angular displacement** is defined as:
  "The change in the rotational position of an object, measured as the angle between its initial and final positions (where the angle can be in radians, degrees or revolutions)
- However, the SI unit for angular displacement is **the radian**
- It is a vector quantity with both magnitude (the angle) and direction (clockwise or counterclockwise) and is typically measured in radians, degrees, or revolutions. The SI unit for angular displacement is the radian, and it can be calculated by dividing the arc length by the radius
- It can be calculated by dividing the arc length by the radius

In essence, Angular displacement is "The angle in radians through which a point or line has been rotated in any given direction about a specified axis"

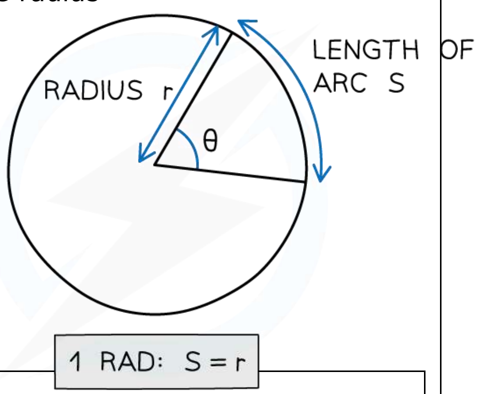
$$s = r\theta$$

linear displacement = radius x angular displacement

$$\Delta\theta = \frac{\text{distance travelled around the circle}}{\text{radius of the circle}}$$

where θ must be in radians, not degrees

1 revolution = 2π rad = 360°



Relation between angular and linear displacement
Angular displacement is related to linear displacement by the arc length, $s = r\theta$
Angular displacement is the change in angular position over time, similar to how linear displacement is the change in position over time

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understand what is meant by *angular velocity* and be able to use the equations $v = \omega r$ and $T = \dfrac{2\pi}{\omega}$

Angular velocity (ω) is defined as:

"**The rate of change of angular displacement** of a rotating body" (in rad/s)

- It is the the rate at which an object rotates or revolves around an axis, measuring how fast its angular position changes over time
- Angular velocity is a vector, meaning it has both magnitude (the rotational speed) and direction (axis of rotation)

$$\omega = d\theta \,/\, dt$$

If the object completes a full circle (2π radians) in a time period, T, then the angular velocity is given by:

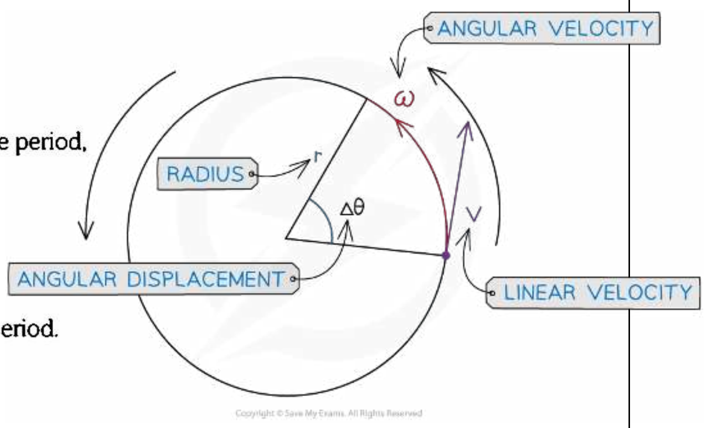
$$\omega = \frac{2\pi}{T}$$

$$\therefore \quad T = \frac{2\pi}{\omega}$$

The frequency of rotation is the reciprocal of the time period.

$$f = \frac{1}{T}$$

$$\therefore \quad \omega = 2\pi f$$



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be able to use vector diagrams to derive the equations for centripetal acceleration $a = \dfrac{v^2}{r} = r\omega^2$ and understand how to use these equations

Mind you that this object moving in uniform circular motion travels with constant angular velocity but changing linear velocity

Centripetal Acceleration

- An object in uniform circular motion moves in a circle at a constant speed but its velocity is not constant as its direction is always changing
- This means the object is always accelerating (as velocity is always changing)
- This acceleration is called **centripetal acceleration**, and it is always directed towards the center of the circle
- The centripetal acceleration is caused by a constant net force, the **centripetal force**, which is also directed towards the center of the circle

This formual derivation is one of the most commonly asked questions in U4!

Derivation of centripetal acceleration using vector diagrams!

By comparing two similar isosceles triangles- *one formed by position vectors and displacement*, and *the other by velocity vectors and the change in velocity*- you can show that the ratio of the change in velocity to the speed is equal to the ratio of the arc length to the radius

$$\frac{\text{change in velocity}}{velocity} = \frac{displacement}{radius}$$

$$\frac{\Delta v}{v} = \frac{\Delta s}{r}$$

When the time interval is small, the change in position is approximately the arc length ($v\Delta t$), leading to the formula $a_c = \dfrac{v^2}{r}$

This can be extended to $a_c = \dfrac{v^2}{r} = r\omega^2$ by substituting $v = r\omega$

https://www.youtube.com/watch?v=mL37CMIjQQ4

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Equations of Centripetal Acceleration

For any particle to move in a circular path, there must be a net force directed towards the centre of the circle to provide an acceleration, known as centripetal acceleration

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understand that a resultant force (centripetal force) is required to produce and maintain circular motion

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Centripetal Force *F* is defined as:

"The resultant force towards the centre of the circle required to keep a body in uniform circular motion that is always directed towards the centre of the body's rotation"

The centripetal force is not a separate force of its own

- It can be any type of force, depending on the situation, which keeps an object moving in a circular path
- Eg: Magnetic force on a charged particle is always centripetal because the force acts at 90° to the charged particle's velocity

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be able to use the equations for centripetal force $F = ma = \frac{mv^2}{r} = mr\omega^2$.

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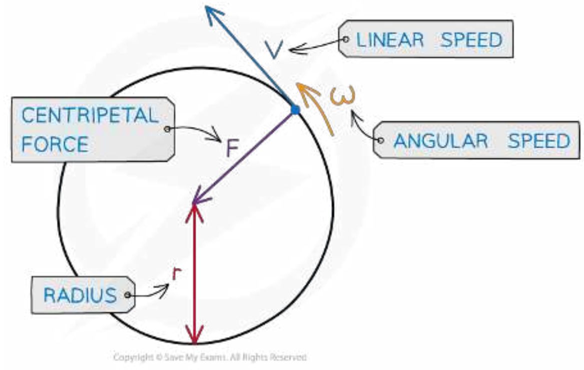
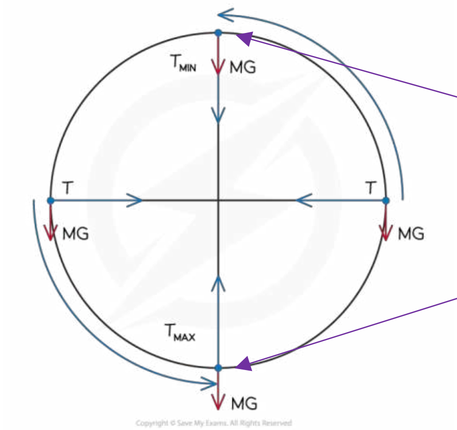
Centripetal Force Equation

$$F = ma = \frac{mv^2}{r} = mr\omega^2$$

The centripetal force is the **resultant** force on the object moving in a circle



Vertical Circular Motion



At the top of the circle,
$T_{min} = mv^2/r - mg$

At the bottom of the circle,
$T_{max} = mv^2/r + mg$

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Remarks

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